

Hariprakash Karthikeyan

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EDUCATION

University of Rochester

Master of Science, Data Science

Aug 2024 - Dec 2025

- **Coursework:** Data Science at Scale, Statistical Machine Learning, Network Science Analytics, Database Systems

SRM Institute of Science and Technology

Bachelor of Technology, Electronics and Communication Engineering

Aug 2020 - May 2024

EXPERIENCE

Buffalo Solar

Data Analyst Intern

Jun 2025 - Jul 2025

Buffalo, New York

- Designed and developed an interactive dashboard using Streamlit and Plotly to monitor and analyze solar energy production across 500+ SolarEdge and CPS sites, visualizing over 5 years of monthly energy and power trends with dynamic filters.
- Automated data ingestion from the SolarEdge API to consolidate over 1,000 CSV files, ensuring real-time performance monitoring and outage detection with customizable thresholds, and maintaining data accuracy.
- Engineered predictive performance analysis by comparing actual versus forecasted energy generation using interactive line and bar plots, and quantified error with MAE and RMSE metrics to support proactive energy planning.
- Facilitated flexible comparison of monthly metrics across multiple years and enabled site-specific prediction file uploads, thereby enhancing analytical efficiency for processing 1000+ data points per site.

University of Rochester Medical Center

Research Assistant- Healthcare Data Science

Aug 2025 - Dec 2025

Rochester, New York

- Constructed forecasting models on continuous physiologic and EHR datasets to address the challenge of traditional scores missing up to 80% of at-risk patients, thereby improving data-driven patient monitoring.
- Developed early warning algorithms to predict cardiorespiratory decompensation (CRD), aiming to elevate survival outcomes for in-hospital cardiac arrest patients beyond the current 25% discharge rate.
- Modeled dynamic vital sign trends and physiologic variability to capture high-risk deterioration patterns, reducing reliance on infrequent manual documentation and enhancing sensitivity in early detection.
- Crafted clinical decision support prototypes aligned with rapid response protocols, contributing to potential decreases in preventable events among the 290,000 annual U.S. in-hospital cardiac arrest cases.

University of Rochester Medical Center

Data Scientist

Oct 2024 - May 2025

Rochester, New York

- Enhanced diagnostic accuracy by 15% using Variational Autoencoders and PCA for feature extraction and antibody detection in retinal tissue images; deployed a deep learning model with Keras and TensorFlow on 5,000 retinal images, achieving 94.3% accuracy in the early diagnosis of Diabetic Retinopathy and ROP.
- Improved data integrity and accessibility by managing and pre-processing clinical data from multiple sources, performing data cleaning, transformation, and validation; designed an SQL database to integrate EPIC system data across three departments, increasing data retrieval speed by 30%.
- Increased forecasting accuracy by 40% through Time Series Analysis to predict patient outcomes and hospitalization trends, delivering actionable insights that reduced diagnostic turnaround time and operational costs by 25%.
- Validated drug efficacy through A/B testing on 50 subjects using statistical tests (t-tests, ANOVA, chi-square), confirming treatment effectiveness and enhancing clinical decision-making.

PROJECTS

Data Science Capstone

Butler-Till

Aug 2025 - Dec 2025

Rochester, New York

- Analyzed 22,000+ rows of time-series media spend and revenue data using hierarchical Bayesian modeling to generate region- and city-level insights, improving accuracy over national-level aggregates
- Delivered data-driven recommendations on media effectiveness by quantifying revenue impact across markets, blending advanced statistical modeling with business-focused analysis to support strategic decision-making

SKILLS

- **Frameworks:** Matplotlib, Numpy, Sci-kit, Pandas, PyTorch, TensorFlow, Keras, HuggingFace, Apache Spark, Hadoop
- **Languages:** Python, R, SQL, HTML, CSS, PHP, MongoDB
- **Tools:** Power BI, Tableau, OpenCV, Microsoft Excel
- **Cloud Platform:** Databricks (on AWS), Azure